

ENCRYPTION AND DIGITAL SIGNATURE

Pre-requisites:

Channel has to generate a pair of keys (Private and Public) and Public Key need to be shared with EIS.

Algorithm Specification:

Advanced Encryption Standard (AES) for Payload Encryption.

- Cipher Mode Operation : Cipher Block Chaining (CBC) with PKCS5Padding
- Cryptographic Key : 256 bits
- IVector : First 16 byte of cryptographic key (Secret Key)

Rivest-Shamir-Adleman (RSA) for AES Key Encryption.

- Cipher Mode Operation : Electronic Codebook (ECB) with OAEPPadding
- Cryptographic Key : 2048 bit X509 Certificate

SHA256-Rivest-Shamir-Adleman (RSA) for Digital Signature.

- Hashing Algorithm : SHA 256
- Cryptographic Key : 2048 bit X509 Certificate

Implementation and Process Flow:

- 1) Channel has to generate a 32 character plain text dynamic key (AES 256 encryption key) for the payload encryption.
- 2) Encrypt the plain JSON request with above AES 256 encryption key.
- 3) SHA256-RSA algorithm has to be used to sign the plain request payload with the help of Channel Private Key.
- 4) RSA algorithm has to be used to encrypt the above AES 256 encryption key with the help of shared EIS Public Key.
- 5) All the three values (obtained from Steps 2, 3, 4) must be in **Base64 encoding** format and shared with EIS at the time of request in the below mentioned format.

```
{  
    "REQUEST_REFERENCE_NUMBER": "SBISI2511190000000000000006",  
    "REQUEST": "[Outcome of Step (2)]",  
    "DIGI_SIGN": "[Outcome of Step (3)]"  
}
```

Add the secret key in the Http Header request

AccessToken - [Outcome of Step (4)]

6) EIS will validate the request received and decrypt the AccessToken with RSA algorithm and then proceed with decryption of REQUEST using the AES encryption key obtained from the decryption of AccessToken and verification of DIGI_SIGN with the shared Channel Public Key.

7) Failure in Validation of request received will return ERROR_CODE **SI011** with **401** HTTP code and the failure in decryption/verification will return ERROR_CODE **SI051** with **401** HTTP code.

```
{
  "REQUEST_REFERENCE_NUMBER": "SBISI2511190000000000011",
  "ERROR_CODE": "SI051",
  "ERROR_DESCRIPTION": "Unable to process due to technical error!!"
}
```

Response HTTP Header

X-Original-HTTP-Status-Code - 401

8) On Successful decryption/verification EIS will move on for valid request processing. Depending on success/failure EIS will forward the response to the Channel with **200** HTTP code.

```
{
  "RESPONSE": "[Encrypted Response (Encrypted with the key obtained from the decryption of AccessToken)]",
  "REQUEST_REFERENCE_NUMBER": "SBISI2511190000000000010",
  "RESPONSE_DATE": "26-11-2019 13:10:17",
  "DIGI_SIGN": "[Signed Data (Signed on Plain Response)]"
}
```

9) Finally channel must decrypt the RESPONSE using the same AES 256 encryption key which was used in for REQUEST and verify the DIGI_SIGN with SHA256 RSA algorithm with shared EIS Public Key.

REFERENCE SCREENSHOTS:

➤ Prepare JSON request using encrypted payload and encrypted key as shown below

The screenshot displays a REST client interface with two panels. The top panel shows the 'Body' tab with a JSON request. The bottom panel shows the 'Headers' tab with two headers: 'Content-Type' and 'AccessToken'. The 'AccessToken' header is circled in red.

Request Body (JSON):

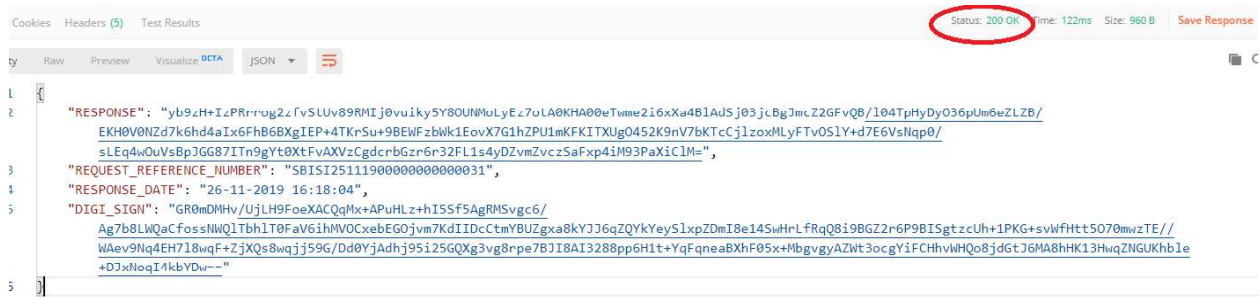
```
{
  "REQUEST_REFERENCE_NUMBER": "SBISI2511190000000000031",
  "DIGI_SIGN": "eLJTe4xxu/TFyuVJf9H6KUCrL4kOmSCEAxICse8k6fbo4Vtzlc+GRbr/gYUujgfgFVMbVCBjXtEMR/BH7Pr0NbMmGiNpKZjph4TiWmVr5EMEhhwGxqvaQxm70/RqdXMaqjln490Fex5I1Uut5mcsSQ5T3RzXGQ2HH4DLtaX8a1Q97S0ESyqA0Zsmr0imNkhVedSEELUNdxmeU4W2Wu+D75vSkky6LFccWdF7INhv/h2swr0p70WhAmRv+B74D77TnVrhvIKBp0B3WGDYKItcDrBrHayhvOY9QkEDpUmrIA6o6yypYmSxUMAZvebZKD8Sf+XCxniIzLuwBtxjnaXZ18hHw==",
  "REQUEST": "ZD+jX7aCqbDntNLkaJwSkmlM53oU6WQ7oCD1nHvbJQBywIjP6GeQRqPCCX8M+p01b/kDQgC+CyJ5SnnkcEGwCREjg7Dd5YDhEhtzon8pJByc7KDRKBnQWuOd+sq+yo3dN0AGGZTETi6+bEzjOHmL+dSEq9MIH2JYOhEm6Jr3XsR3J1LzRBegGDzUHh4suW0f4ZcuXq5bA2LeqQ7Ud7jnYt1nXcbzgKqLwUff/126xtC3YFY1MrB1HCj/xW4L9B9thA27hfByzQeR/CeogA=="
}
```

Headers:

KEY	VALUE	DESCRIPTION
Content-Type	application/json	
AccessToken	QuF2NhdTjy+8uVQwT1clM55UCm6CSxUjiOpO7Pv7VtwQjV/nLJfEVQ0fw7juxRmKQuu chLTM74wHZ4UZ8QbKUudRmEgN5zkTQ+RofjcW4GDfr4osQyZ/3pHDGf40UTTwpR cBbl3rmuTZwdYfYz1qr4BGIYKNNw/Dq0CFT07NxmGMZyAWuX1JfV8cVHmD95jmbbg9 eNAaxETO3hROYEmWPFMy+sLY3iXlgb11AD0i3vErYzpHLiKBs4bn2TbTYL+KARZ63k1 u/cDQ5tCEyj2/7wqj8X4b+dZoCk5hFM9xNxyWnMBzFzKFWAj6SKBiFZjo27fhckCLWX 11pWaknq9lvw==	Description

NOTE: - Payload is passed in body as *REQUEST* field, Signature is passed in body as *DIGI_SIGN* field and encrypted key in header as *AccessToken*.

➤ Encrypted response format after successful processing of the request.



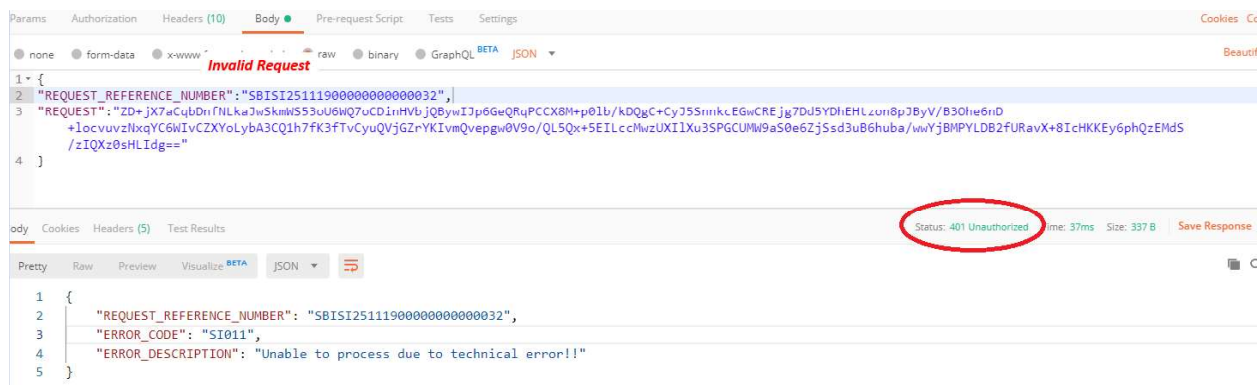
```
1 {
2   "RESPONSE": "yb9zH+IzPrRrrug2zfvSUVv89RMIj0vuiky5Y80UNMoLyEz7oLA0KHA00eTwme2i6xXa4B1AdSj03jcBg3mcZ2GFvQB/104TpHyDyO36pUm6eZLZB/
3   EKHOV0NZd7k6hd4aIx6fH8B8XgIEP+4TKrSu+9BEWFzbWk1EovX7G1hZPU1mFKITXUg0452K9nV7bKTCcjlzoxMLyFTv0S1Y+d7E6VsNqp0/
4   sLEq4wOuVs8pJGG87ITn9gYt0XtFvAXVzCgdcrbGzr6n32FL1s4yDZvmZvczSaFxp4iM93PaXiC1M=",
5   "REQUEST_REFERENCE_NUMBER": "SBISI25111900000000000031",
6   "RESPONSE_DATE": "26-11-2019 16:18:04",
7   "DIGI_SIGN": "GR0mDMHv/UjLH9FoeXACQqMx+APuHLz+hI55F5AgRMSvgc6/
8   Ag7b8LWQaCfossNWQ1Tbh1T0faV6ihMVOcxebEG0jvm7KdIIDcCtmYBUZgxa8kYJ36qZQYkYeyS1xpZ0mI8e14SwHrLFRQ8i9BGZ2r6P9BISgtzcUh+1PKG+svwFht5070mwzTE//
9   WAAev9Nq4EH718wqF+ZjXQs8wqj59G/Dd0YjAdhj95i25GQXg3vg8rpe7B3I8AI3288pp6H1t+YqFqneaBXhF05x+MbgyvgyAZWt3ocgYiFCHhvWQo8jdGtJ6MA8hHK13HwqZNGUKhb1e
10  +DJxNoqI4kbYDw--"
11 }
```

➤ Authorization/Authentication Failure response format before processing the request.



```
1 {
2   "REQUEST_REFERENCE_NUMBER": "SBISI25111900000000000032",
3   "ERROR_CODE": "SI051",
4   "ERROR_DESCRIPTION": "Unable to process due to technical error!!"
5 }
```

➤ Validation Failure response format before processing the request.



```
1 {
2   "REQUEST_REFERENCE_NUMBER": "SBISI25111900000000000032",
3   "REQUEST": "ZD+ jX7aCubDn1NLkaJwSkmlWS53uU6WQ7uCD1nHVBjQBywI3p6GeQRqPCCX8M+p01b/kDQgC+Cy75SrmkLEgwCREjg7DU5YDhEHLzoni8p7ByV/B30Ihe6rD
4   +locvuvzNlxqYC6WIVCZXYoLybA3CQ1h7fK3ftvCyUQVjGZrYKIVmQvrepw0V9o/QL5Qx+5EILccMwzUXI1Xu3SPGCUmM9a50e6Zjsd3uB6huba/wvYjBMPYLD82fURavX+8IcHKKEy6phQzEMdS
5   /zIQxz0sHLIdg=="
6 }
```